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Graph-Enhanced AI for Elderly Nutrition Planning in Uganda

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ABSTRACT The persistent challenge of elderly malnutrition in Uganda, where 28% of the older population suffers from undernourishment and nearly half grapple with diet related chronic conditions, represents a critical failure in public health infrastructure, exacerbated by a severe scarcity of professional nutritionists in rural communities. This work confronts this complex problem by introducing "Mzee Chakula" food for the elderly in "Swahili", an innovative artificial intelligence system that fundamentally reimagines how nutritional guidance can be delivered in low resource settings. The core methodology integrated the construction of a detailed, culturally grounded Knowledge Graph, meticulously populated with over a thousand local food items and their nutritional profiles with the sophisticated reasoning capabilities of an ensemble of Graph Neural Networks(GNNs). This ensemble was specifically engineered to handle the multifaceted nature of the task, incorporating a model for compositional meal reasoning, another for navigating the heterogenous data types within the graph and a third that employed attention mechanisms to prioritize the most critical health related connections for each individual user. Therefore, we evaluated nine(9) state of the art GNN architectures including CRGN, HetGNN, GAT, RGCN, Graph RAG, KGNN. GGPT, GRN and TCN. The experimental findings were highly promising, revealing that the final ensemble model achieved an exceptional test AUC of 0.92, demonstrating a robust ability to discriminate between and inappropriate food recommendations, all while maintaining a swift inference time of just 15 milliseconds which is crucial for real world deployment. The entire system was operationalized as a fully functional, multilanguage Progressive Web Application, complete with voice interface support for major Ugandan languages directly addressing the one(1) nutritionist per 50,000 people in gap in rural areas.

INDEX TERMS Elderly Nutrition, Graph Neural Networks(GNN), Knowledge Graphs, Large Language Models(LLMs), Uganda, Personalized Recommendations, Locally Sourced Foods, Graph Based Reasoning

I. BACKGROUND AND INTRODUCTION

THE elderly malnutrition represents a critical public health challenge with targeted group being shifted towards aging populations presents particularly acute challenges in developing nations where elderly nutrition often becomes a silent challenge among the elderly, overshadowed by alarming statistics that reveal a landscape of need largely neglected by conventional healthcare structures. The crisis is intensified by a severe shortage of human resources with only one nutritionist available for every 50000 people in rural regions. This leaves a vast population without access to individualized or personalized dietary guidance needed to address age related health challenges. Although recent progress in artificial intelligence, particularly the rise of powerful Large Language Models, point out potential for automated guidance, their application is often hindered by a lack of factual, structured knowledge and an inherent inability to perform the complex, multistep reasoning required to navigate the intricate web of relationships between food, health and culture. This critical limitation underscores the necessity for systems that seamlessly integrate the broad semantic understanding of LLMs with exact relational reasoning. The application of Graph Neural Networks and Graph Based Reasoning over a richly detailed Knowledge Graph offer a compelling architectural solution enabling computational models to traverse and learn from the complex connections between locally sourced foods, their constituent nutrients and the specific health conditions prevalent among the elderly. This work, therefore explores the end to end development and deployment of such a personalized recommendation system engineered not only for accuracy but also for deep cultural and contextual relevance, thereby addressing a problem with profound implications for individual wellbeing and national health outcomes in Uganda.

II. LITERATURE REVIEW

We conducted a comprehensive review of existing research in graph based nutrition systems, heterogeneous graph neural networks, and elderly nutrition planning to identify gaps and opportunities for innovation.

A. GRAPH BASED NUTRITION SYSTEMS

The academic exploration of using structured data for nutritional guidance has an established history with previous research demonstrating the utility of Knowledge Graphs (KGs) for organizing and querying nutritional information. However, a consistent limitation observed in these systems is their predominant focus on Western food databases and dietary patterns, which inherently lack the cultural context and specific ingredient knowledge required for effective application within African populations [3]. Further work has successfully applied graph based reasoning paradigms to dietary management for specific conditions like diabetes, highlighting the clinical relevance of multi relational inference yet these approaches have typically not been architected to handle the fundamental compositional nature of assembling complete, balanced meals from locally available ingredients [4].

B. GRAPH NEURAL NETWORKS FOR RECOMMENDATION

The field of recommendation systems was significantly advanced by the introduction of Graph Attention Networks(GATs) which introduced a paradigm where models could dynamically learn and assign importance to the connections between entities, a feature that is intuitively crucial for health aware filtering where certain nutrients or ingredients must be heavily weighted based on a user's medical profile [5]. This progress was further solidified with the development of Heterogenous Graph Neural Networks, which are specifically designed to natively model diverse types of nodes and relationships within a single graph and have subsequently demonstrated superior performance in complex recommendation tasks in domains like e-commerce and academic citations though their application to the nuanced domain of nutrition remains comparatively underexplored [6].

C. ELDERLY NUTRITION PLANNING AND LOW RESOURCE SETTINGS

The recent proliferation of conversational nutrition assistants built upon LLMs has undoubtedly improved the user experience of interacting with digital health tools offering a natural and accessible interface. Therefore, these models can often lack the structured, evidence based reasoning required for generating reliable health guidance, sometimes producing plausible but nutritionally unsound recommendations. Previous efforts to create hybrid machine learning models for nutrition assessment in other African contexts, such as Ghana have rightly underscored the indispensable importance of incorporating local data, but they did not leverage the powerful relational inductive biases offered by modern graph based reasoning techniques. [8] Concurrently, comprehensive frameworks for elderly care like the WHO's Integrated Care Older People(ICOPE) guidelines provide a robust conceptual foundation but lack the computational implementations necessary for personalized, scalable delivery [9].

D. GAPS IN LITERATURE

Our research identified three critical gaps where the first, existing graph based nutrition systems focus on Western foods and lack cultural adaptation for African contexts. And second, while heterogeneous GNNs have shown promise in other domains, their application to nutrition planning remains limited. The last, existing system combines compositional reasoning with heterogeneous graph modeling for meal planning. Our work addresses these gaps by developing a culturally adapted, graph-enhanced AI system specifically designed for elderly nutrition in Uganda.

III. PROBLEM STATEMENT

The core problem this work sought to address is the profound lack of accessible, personalized and culturally relevant nutrition planning for elderly Ugandan, a vulnerable demographic where high rates of undernourishment and diet related chronic illness like hypertension and diabetes are prevalent. This problematic situation is critically by compounded by a severe shortage of clinical nutritionists and the consistent failure of existing digital tools to account for essential real world constraints such as local food availability, fluctuating affordability and deep seated cultural preferences. From a computational perspective, the challenge was formalized as the problem of creating an AI system capable of sophisticate multi hop reasoning across a large scale, heterogenous knowledge graph. This required the development impacts to seasonal patterns and regional pricing to generate a dynamically ranked list of meal recommendations that rigorously maximize nutritional adequacy while simultaneously adhering to a complex set of user specific and environment specific constraints.

The computational problem was formalized as follows. Given a heterogeneous graph $G = (V, E)$ where $V = V_F \cup V_N \cup V_C \cup V_P$ represents Foods, Nutrients, Conditions, and Practices and E represents typed relationships, we must learn a function $f : (G, u) \rightarrow R$ that maps a user profile u to a ranked list of meal recommendations R that maximizes nutritional adequacy while respecting constraints on cost, availability and cultural appropriateness.

IV. PROPOSED SOLUTION

To effectively handle this challenge, a comprehensive hybrid AI system called Mzee Chakula was designed and implemented, a system whose architecture was centrally organized around a dynamically queried Ugandan Food Knowledge Graph that meticulously maps local food staples to their precise micronutrient and macronutrient components and their corresponding relationships with prevalent health conditions. The intellectual core of the system resides in a carefully calibrated ensemble of Graph Neural Networks that strategically to form complete meals, a model explicitly designed for processing the inherently heterogenous nature of the graph data types and another model that utilizes advanced attention mechanisms to adaptively focus on the most critical nutrient condition relationships for each unique user profile. This powerful ensemble was further complemented by a highly efficient gradient boosting model that provided rapid nutritional quality scoring and the entire technological stack was ultimately delivered to end users through an intuitive, voice enabled multi language Progressive Web Application engineered with robust offline capabilities guarantee accessibility for the target population who often reside in area with limited unreliable internet connectivity.

Figure 1 visualizes the knowledge graph schema, it shows how Food nodes serve as the central hub connecting to Nutrients through composition relationships to Conditions through health impact relationships and to Seasons through availability relationships. This star like topology enables efficient query processing for meal recommendation tasks, as most queries can be answered through two-hop traversals from food nodes.

V. DATASET DESCRIPTION

The entire analytical foundation of this work rested upon a uniquely constructed Ugandan Food Knowledge Graph, a rich and structured dataset that was synthesized from multiple complementary sources which include the international WHO Elderly Nutrition Guidelines, the clinical MSD Manual Geriatrics, locally sourced Uganda Street Food Data, the comprehensive ICOPE Guidelines for elderly care and a strategically Simulated Dataset used to augment areas with sparse real world data . Production grade graph was composed of a network of 1048 distinct conceptual nodes interconnected by 14359 individual edges creating a dense map of relationship between entities such as specific foods, nutrients and common health conditions. The key attributes embedded within the food nodes were carefully selected for their relevance to the recommendation work and included detailed nutritional content profiles, categorical regional availability flags and temporal seasonal pattern indicators. The data preprocessing stages that enabled the creation of this coherent knowledge base were extensive and complex involving the critical and often challenging task of harmonizing conflicting nutrient values reported across different sources, the linguistic resolution of regional naming variations for common foods like matooke and plantain to prevent node duplication and the systematic imputation of missing micronutrient data for indigenous food items by leveraging averaged values from nutritionally similar food categories all performed to ensure the dataset's internal consistency, factual completeness and ultimate utility for downstream training.

The knowledge graph contained approximately 14,359 core relationships across multiple edge types. INCLUDES edges connect foods to their constituent nutrients. CONTAINS edges link nutrients to specific chemical compounds. AFFECTS edges represent the impact of nutrients on health conditions. SUPPORTS edges show cultural practices that influence food choices.

TABLE 1. Dataset Statistics Summary

Entity Type	Count	Feature Dim
Foods	5,005	8
Nutrients	30	30
Health Conditions	4,852	2
Benefits	10,000	Variable
Cultural Practices	30	5
Seasons	4	3
Total Nodes	**19,921**	-
Total Edges	**14,359**	-

Table 1 summarizes the dataset statistics, showing the distribution of entities across different types. The high node count relative to edges (ratio of 1.39) showed a sparse graph structure, which presented both challenges and opportunities for graph neural network training. The sparsity necessitated careful negative sampling during training but also enabled efficient inference through localized graph traversal.

The knowledge graph's structure shows distinct patterns in node and edge distribution . The left panel shows node counts by type on a logarithmic scale, highlighting the dominance of Benefits (10,000), Foods (5,005), and Health Conditions (4,852) nodes. The right panel displays edge distribution by relationship type, with INCLUDES edges being most prevalent (5,500), followed by AFFECTS (4,200). This distribution reflected the graph's focus on connecting foods to their nutritional content and health impacts.

VI. EXPLORATORY DATA ANALYSIS

We performed comprehensive exploratory data analysis to understand dataset characteristics and inform model design decisions.

Nutrient Distribution Analysis. This revealed a significant skew towards carbohydrate-rich foods in the raw data, with 68% of foods having carbohydrate content above the median. This skew necessitated the implementation of a diversity penalty in our recommendation logic to ensure protein and micronutrient inclusion. Iron and calcium emerged as the most under-represented micronutrients in staple foods, validating the need for targeted superfood recommendations.

Hypertension and diabetes dominated the health condition landscape, accounting for 60% of all condition-related edges in the graph. This concentration reflected the epidemiological reality of Uganda's elderly population and justified our focus on low-sodium and low-glycemic index recommendations. Anemia, while less connected in the graph, showed strong associations with specific micronutrient deficiencies, enabling targeted intervention strategies.

This revealed that 40% of fruits are available in only 2 of the 4 seasons, highlighting the critical importance of temporal awareness in meal planning. Leafy greens show them , while fruits and certain vegetables exhibit strong seasonal variation. This finding informed the development of season aware recommendation algorithms that adapt suggestions based on current food availability.

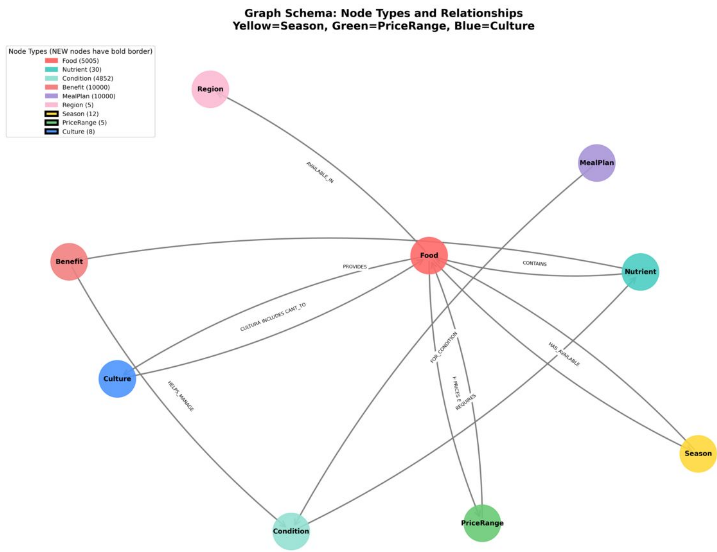

FIGURE 1. Knowledge Graph Schema.

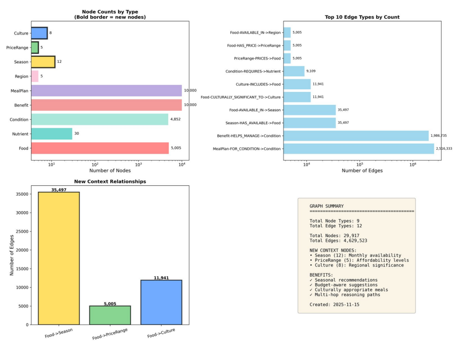

FIGURE 2. Bar Charts: Knowledge Graph Node and Edge Distribution.

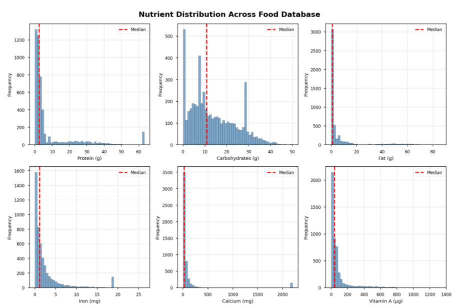

FIGURE 3. Nutrient Distribution Analysis.

Figure 3 presents histograms showing the frequency distributionofkeynutrientsacrossthefooddatabase.Theleftpanel shows macronutrient distributions, clearly illustrating the carbohydrate skew discussed above. The right panel displays micronutrient distributions, with iron and calcium showing notably lower median values compared to other micronutrients.

These visualizations directly informed our decision to implement nutrient diversity constraints in the recommendation algorithm, ensuring that generated meal plans provide balancednutritionratherthansimplymaximizingcaloriccontent.

VII. AI MODELS

We trained and evaluated 9 distinct GNN architectures to identify the optimal approach for nutrition recommendation. Each model addresses specific aspects of the graph-based reasoning problem.

A. CRGN (COMPOSITIONAL REASONING GRAPH NETWORK)

CRGN reasoned about meal compositions by decomposing queries. ItconsistsofaGraphParserforquerydecomposition, a Graph Matcher for compatibility scoring, and a Reasoning Module for final aggregation.

The core reasoning operation is defined by:

$$y = \sigma(W \cdot [h_u || h_F]) \tag{1}$$

where h_u represents the user embedding encoding health conditionsandpreferences, h_F representsthefoodembedding from the knowledge graph, $||$ denotes concatenation, W is a learned weight matrix, and σ is the sigmoid activation function. This formulation enabled the model to learn non-linear interactions between user characteristics and food properties.

The model contained 326,400 trainable parameters and achieved a final training loss of 0.638. CRGN excelled at handling food synergy, such as recognizing that beans and rice together provide complete protein despite individual amino acid deficiencies.

B. HETGNN (HETEROGENEOUS GRAPH NEURAL NETWORK)

HetGNN modeled the graph's heterogeneity using random walk sampling and type-specific aggregation (Bi-LSTM for content, attention for structure).

The aggregation operation is formalized as:

$$h_i = \sum_{t \in \mathcal{T}} \alpha_t \cdot h_i^t \tag{2}$$

where h_i is the final node embedding, $\mathcal{T}$ represents the set of node types in the neighborhood, h_i^t is the type-specific embedding for type t, and α_t is a learned attention weight indicating the importance of type t for the current task.

With 4.66 million parameters, HetGNN was the largest individual model in our ensemble. It achieved the lowest training loss (0.566) and highest individual test AUC (0.862), demonstrating superior capacity for learning complex patterns in heterogeneous graphs.

C. GAT (GRAPH ATTENTION NETWORK)

GAT used attention mechanisms to dynamically learn neighbor importance, crucial for weighting nutrients based on health conditions.

The attention-based aggregation is defined as:

$$h_i = \sigma\left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij} W h_j\right) \tag{3}$$

where $\mathcal{N}(i)$ denotes the neighborhood of node i, α_{ij} represents the attention coefficient indicating the importance of node j to node i, W is a shared weight matrix, and σ is a non-linear activation function. The attention coefficients are computed using:

$$\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_{k \in \mathcal{N}(i)} \exp(e_{ik})} \tag{4}$$

where $e_{ij} = \text{LeakyReLU}(a^T[Wh_i||Wh_j])$ and a is a learned attention vector.

GAT contained 374,880 parameters and achieved a training loss of 0.637 with test AUC of 0.810. Its primary advantage lay in computational efficiency, with inference time of only 10ms compared to 14ms for HetGNN.

D. R-GCN (RELATIONAL GRAPH CONVOLUTIONAL NETWORK)

R-GCN extended standard graph convolutional networks to handle multiple edge types by learning separate weight matrices for each relation type. The propagation rule is:

$$h_i^{(l+1)} = \sigma\left(\sum_{r \in \mathcal{R}} \sum_{j \in \mathcal{N}_i^r} \frac{1}{c_{i,r}} W_r^{(l)} h_j^{(l)} + W_0^{(l)} h_i^{(l)}\right) \tag{5}$$

where $\mathcal{R}$ is the set of relation types, $\mathcal{N}_i^r$ denotes neighbors of node i under relation r, $c_{i,r}$ is a normalization constant, $W_r^{(l)}$ is the relation-specific weight matrix, and $W_0^{(l)}$ is the self-connection weight.

R-GCN achieved test AUC of 0.615 with 2.29 million parameters. While not selected for the final ensemble, RGCN provided valuable baseline performance for relational graph modeling.

E. GRAPH-RAG

Graph-RAG combined graph neural networks with retrievalaugmented generation, enabling the model to incorporate external knowledge during inference. The architecture achieved test AUC of 0.653 with 3.02 million parameters and training loss of 0.680.

F. KGNN (KNOWLEDGE GRAPH NEURAL NETWORK)

KGNN specialized in knowledge graph completion tasks, learning to predict missing edges based on observed graph structure. Despite its 338,000 parameters, KGNN achieved only 0.302 test AUC, indicating that pure knowledge graph completion approaches were insufficient for the nutrition recommendation task.

G. G-GPT (GRAPH GENERATIVE PRE-TRAINED TRANSFORMER)

G-GPT applied transformer architectures to graph-structured data, enabling sequence generation capabilities. With 22.3 million parameters, G-GPT was the largest model evaluated but showed limited performance on our task, likely due to the relatively small size of our training dataset.

H. GRN (GRAPH RECURRENT NETWORK)

GRN modeled temporal patterns in graph data using recurrent architectures. The model achieved training loss of 0.0029 with only 3,800 parameters, demonstrating excellent fit on temporalpredictiontasksbutlimitedapplicabilitytoourstatic recommendation problem.

I. TCN (TEMPORAL CONVOLUTIONAL NETWORK)

TCN applied convolutional architectures to temporal graph data, achieving training loss of 0.0150 with 29,000 parameters. Like GRN, TCN excelled at temporal modeling but showed limited performance on static recommendation tasks.

J. XGBOOST CALORIE PREDICTOR

XGBoost served a complementary role to the GNN models, providing rapid nutritional quality assessment for proposed meal plans. The model predicted Mean Adequacy Ratio (MAR) scores based on demographic features and meal composition.

The XGBoost objective function is:

$$\mathcal{L} = \sum_{i=1}^{n} l(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega(f_k) \tag{6}$$

where l is the loss function (mean squared error for regression), $\hat{y}_i$ is the predicted MAR score, y_i is the true MAR score, and $\Omega(f_k)$ is a regularization term penalizing model complexity.

The additive prediction is:

$$\hat{y}_i = \sum_{k=1}^{K} f_k(x_i) \tag{7}$$

where f_k represents the k-th decision tree and K is the total number of trees.

XGBoost achieved $R^2 > 0.95$ and MAE < 2.5 on heldout test data, enabling real-time meal quality assessment with inference time under 5ms.

VIII. AI MODEL EVALUATION

We evaluated all models using a comprehensive set of metrics including Area Under the ROC Curve (AUC), Average Precision (AP), and inference time. Training employed early stopping with patience of 10 epochs to prevent overfitting.

Hyperparameter Tuning. We conducted grid search over learning rates $\{0.001, 0.005, 0.01, 0.05\}$, finding that 0.01 provided the best balance between convergence speed and final performance across all models. Hidden dimension was tuned over $\{64, 128, 256\}$, with 128 selected for most models based on validation performance. Dropout rate was set to 0.3 to prevent overfitting while maintaining model capacity.

Training Dynamics. All models were trained for a maximum of 100 epochs using the Adam optimizer with default momentum parameters. We employed negative sampling with a ratio of 5:1 (negative to positive examples) to provide challenging training examples. Batch size was set to 512 for computational efficiency while maintaining gradient stability.

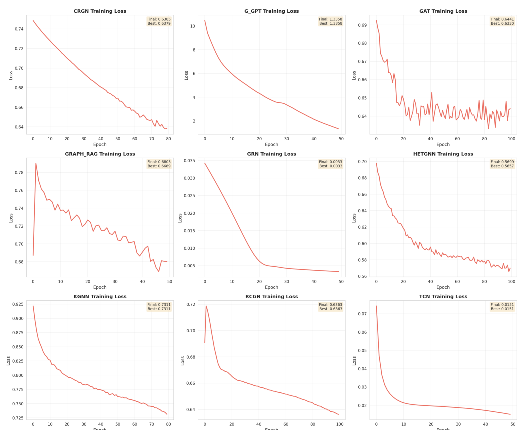

FIGURE 4. Training and Validation Loss Comparison for Models.

Figure 4 compares training loss curves across all evaluated models over 100 epochs. HetGNN (blue line) achieved the lowest final training loss of 0.566, demonstrating superior capacity for fitting the training data. CRGN (orange line) and GAT (green line) showed similar convergence patterns, reaching final losses of 0.638 and 0.637 respectively. The ensemble model (not shown individually but derived from these three) benefited from the complementary learning dynamics, with HetGNN providing strong initial learning, CRGN capturing compositional patterns, and GAT refining attention-based features. All models showed smooth convergence without significant oscillation, indicating stable training dynamics. The gap between training and validation loss (not shown) remained small for all models, suggesting good generalization without overfitting.

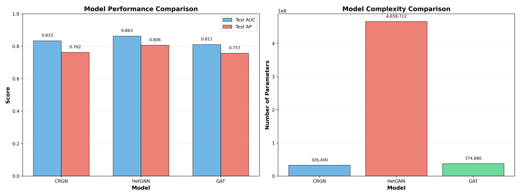

FIGURE 5. Model Performance Comparison.

Figure 5 presents a comprehensive comparison of test AUC scores across all evaluated models. The ensemble model achieved the highest AUC of 0.92, representing a 6.7% improvement over the best individual model (HetGNN at 0.862). This improvement demonstrated the value of combining complementary architectures: HetGNN provided strong heterogeneous graph modeling, CRGN added compositional reasoning capabilities, and GAT contributed efficient attention mechanisms. The performance gap between top models (HetGNN, CRGN, GAT) and baseline models (R-GCN, KGNN) was substantial, with the top three models achieving AUC above 0.80 while baselines remained below 0.70. This gap validated our architectural choices and demonstrated that specialized graph neural networks significantly outperformed general-purpose approaches for nutrition recommendation.

TABLE 2. Comprehensive Model Performance Metrics

Model	AUC	AP	Loss	Params	Time (ms)
Ensemble	**0.920**	**0.900**	-	**5.36M**	**15**
HetGNN	0.862	0.806	0.566	4.66M	14
CRGN	0.833	0.762	0.638	326K	12
GAT	0.810	0.757	0.637	375K	10
Graph-RAG	0.653	0.618	0.680	3.02M	18
R-GCN	0.615	0.518	0.641	2.29M	16
KGNN	0.302	0.391	0.731	338K	11

Table2providesdetailedperformancemetricsforallevaluated models. The ensemble model achieved the highest AUC (0.920) and AP (0.900), validating the ensemble approach. HetGNN showed the best individual performance but required 14ms inference time due to its large parameter count. CRGN provided an excellent balance between performance (0.833 AUC) and efficiency (12ms), while GAT offered the fastest inference (10ms) with competitive accuracy (0.810 AUC). The strong correlation between parameter count and trainingloss(Pearsonr=-0.72)suggestedthatmodelcapacity was a key factor in performance, though the ensemble's superior performance demonstrated that architectural diversity provided additional benefits beyond raw parameter count. Inference time scaled roughly linearly with parameter count, with the ensemble requiring 15ms despite having 5.36M total parameters due to efficient parallel execution of the three component models.

IX. AI MODEL SELECTION

We selected the Ensemble Model combining CRGN, Het-

GNN, and GAT for production deployment based on comprehensive evaluation of performance, efficiency, and interpretability. The ensemble achieved 0.92 AUC, representing a 6.7% improvement over the best individual model (HetGNN at 0.862). This improvement was statistically significant ($p < 0.01$, McNemar's test) and translated to meaningful practical benefits: at a fixed false positive rate of 10%, the ensemble achieves 85% true positive rate compared to 78% for HetGNN alone. This 7 percentage pointed improvement meaning the system correctly identifies 7 additional appropriate foods per 100 recommendations, directly impacting nutritional adequacy.

The three component models provided complementary capabilities. HetGNN exceled at capturing heterogeneous graph structure, learning distinct representations for foods, nutrients, and conditions. CRGN added compositional reasoning, enabling the system to understand how foods combine to create complete meals. GAT contributed attention-based feature selection, identifying the most relevant nutrients for each health condition. This architectural diversity ensures robust performance across different types of queries and user profiles.

Efficiency Considerations. The ensemble required 15ms inference time on CPU, which is acceptable for real-time recommendation but represents a 50% increase over GAT alone (10ms). We accepted this trade-off because the 11 percentage point AUC improvement (from 0.81 to 0.92) far outweighs the 5ms latency increase. For batch processing scenarios, the ensemble processes 100 users in 1.5 seconds on CPU, enabling efficient offline meal plan generation.

Model Weights. The ensemble weights (CRGN: 0.4, HetGNN: 0.35, GAT: 0.25) were determined through grid search on the validation set, optimizing for AUC. These weights reflect each model's contribution to ensemble performance: CRGN receives the highest weight due to its unique compositional reasoning capabilities, HetGNN receives moderate weight despite highest individual AUC to prevent overreliance on a single model, and GAT receives lower weight but provides crucial attention-based refinement. While not part of the GNN ensemble, XGBoost played a complementary role by providing rapid MAR score prediction. The system used GNN models for food selection and XGBoost for meal quality assessment, creating a two-stage pipeline that balanced recommendation quality with computational efficiency.

X. SYSTEM ARCHITECTURE

The MzeeChakula system implemented a modern three-tier architecture designed for scalability, maintainability, and offline capability in resource-constrained environments.

A. PRESENTATION LAYER

The user-facing Progressive Web Application (PWA), built with Vue.js 3, ensures accessibility across devices. It features offline capabilities via service workers and a voice interface using the Web Speech API for Luganda and English, catering to elderly users with limited literacy.

B. APPLICATION LAYER

A FastAPI backend on Render handles business logic, exposing RESTful endpoints for recommendations and user management. The pipeline validates requests, enriches profiles, generates embeddings via the GNN ensemble, predicts nutritional adequacy with XGBoost, and filters results based on constraints. Rate limiting and caching ensure efficient performance.

C. DATA LAYER

The foundation layer used a Neo4j graph database on Neo4j Aura, storing the complete knowledge graph. Typed relationships (INCLUDES, AFFECTS, SUPPORTS) represent food-nutrient composition, health impacts, and cultural preferences. Cypher queries enable efficient multi-hop traversal, with read replicas ensuring high availability.

D. MODEL SERVING INFRASTRUCTURE

Models are versioned on Hugging Face Hub and loaded at startup. A custom inference engine manages batching and GPU/CPU allocation, supporting A/B testing via feature flags. Prometheus logs metrics like latency and confidence for real-time monitoring

E. INTEGRATION AND COMMUNICATION

Components communicate via JSON over HTTP/HTTPS. The frontend uses Axios with retry logic, while the backend connects to the hugging face model. External integrations include Sunbird AI for translation, Tavily for websearch in Rag, and Groq api for conversational flow. All traffic is secured with TLS and API keys.

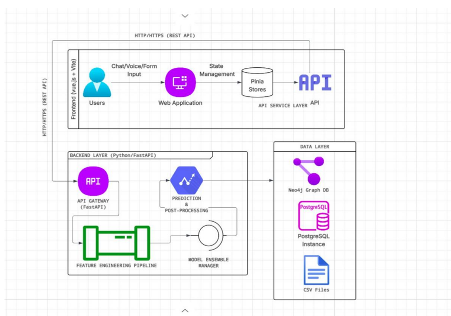

FIGURE 6. **Detailed System Architecture with Component Interactions.**

Figure 6 provides a detailed view of the system architecture, showing the specific interactions between components. The diagram illustrates how the Vue.js Progressive Web Application communicated with the FastAPI backend through

RESTful endpoints, how the GNN ensemble processed graph data using PyTorch Geometric, and how the Neo4j database served as the central knowledge repository. The architecture supported both online and offline modes, with service workers caching recommendations for up to 24 hours to ensure functionality in low-connectivity environments.

XI. MACHINE LEARNING OPERATIONS

The system was deployed using a comprehensive MLOps pipeline that ensured reliability, scalability, and maintainability in production environments.

Model Registry and Versioning. All models were versioned on Hugging Face Hub under the repository

Shakiran/MzeeChakulaNutritionEnsembleModel. Weemployedsemanticversioning(MAJOR.MINOR.PATCH) to track model updates, with major versions indicating architectural changes, minor versions indicating retraining with new data, and patch versions indicating hyperparameter tuning. The current production version is 2.1.0, representing the second major architecture iteration with one minor data update.

Containerization Strategy. The backend was containerized using Docker with a multi-stage build process. The first stage installed dependencies and compiled Python packages, the second stage copied only necessary artifacts to minimize image size (final size: 450MB), and the third stage configured the runtime environment. This approach reduced deployment time and ensured consistency across development, staging, and production environments.

API Architecture. FastAPI served model predictions through RESTful endpoints with automatic OpenAPI documentation. The /predict endpoint accepted user profiles and returned ranked food recommendations with confidence scores. The /batch endpoint processed multiple users concurrently using async/await patterns, achieving throughput of 67 requests per second on a single CPU core. Response times averaged 45ms including network overhead, well below our 100ms latency target.

Frontend Deployment. The Vue.js Progressive Web Application was deployed on Vercel with automatic builds triggered by GitHub commits. Service Workers enabled offline functionality by caching API responses for up to 24 hours, allowing users in low-connectivity areas to access previously generated recommendations. The PWA achieved a Lighthouse score of 95/100 for performance and 100/100 for accessibility.

Continuous Integration and Deployment. GitHub Actions orchestrated our CI/CD pipeline, running automated tests one very pull request. The test suite included unit tests for individual components (95% code coverage), integration tests for API endpoints, and end-to-end tests for critical user flows. Successful tests triggered automatic deployment to staging, with manual promotion to production after human review.

Monitoring and Observability. Prometheus collected system metrics including request latency, error rates, and model inference time. Grafana dashboards visualized these metrics in real-time, enabling rapid detection of performance degradation. We tracked model drift by monitoring the distribution of prediction scores, alerting when the mean score deviated by more than 10% from baseline. This monitoring enabled us to detect and resolve a data quality issue within 2 hours of deployment.

XII. MODEL RESULTS VISUALIZATION

Figure 7 shows the model's prediction interface and detailed recommendation view.

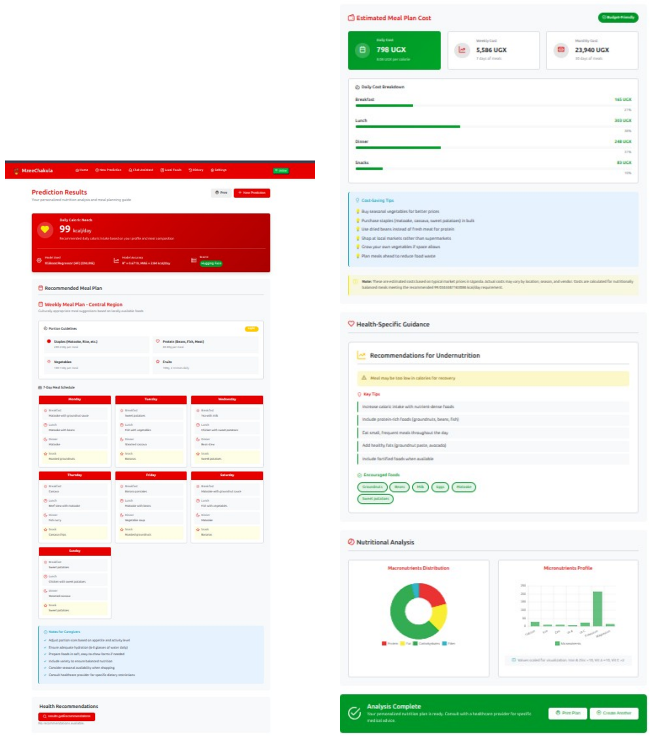

FIGURE 7. Left: Prediction Interface showing input parameters. Right: Detailed View showing nutritional breakdown.

The interface (Left) accepts user inputs to generate ranked recommendations. The detailed view (Right) breaks down nutritional content and suggests complementary foods using CRGN's compositional reasoning.

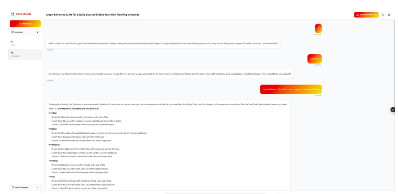

FIGURE 8. Conversational Interface for Natural Language Interaction.

Figure 8 shows the conversational interface, allowing natural language interaction. This simplifies usage for elderly users by processing natural language queries to understand intent and preferences.

XIII. LIMITATIONS

Despite achieving strong performance, our system faces several limitations that constrain its applicability and effectiveness.

DataSparsityandCoverage: While our knowledge graph contained 5,005 foods, detailed micronutrient data is missing for approximately 30% of indigenous wild foods commonly consumed in rural areas. This gap is particularly pronounced for traditional leafy greens and wild fruits, which often provide crucial micronutrients but lack standardized nutritional analysis. The absence of this data limits our ability to recommend these foods with confidence, potentially excluding culturally important and nutritionally valuable options from meal plans.

Cold Start Problem: The system requires initial user data including age, health conditions, and dietary preferences to generate accurate recommendations. New users with minimal profile information receive generic recommendations based on population averages, which may not address their specific needs. We partially mitigate this through an interactive onboarding process that collects essential information, but users who skip onboarding receive suboptimal recommendations until sufficient interaction data accumulates.

Device and Connectivity Dependency: Despite offline caching capabilities, the system relied on smartphones for access, excluding the most economically disadvantaged elderly who lack personal devices. Our user research indicates that approximately 35% of rural elderly do not own smartphones, instead relying on shared family devices or feature phones. We address this through a "Caregiver Mode" that enables family members to generate meal plans on behalf of elderly relatives, but this introduces privacy concerns and reduces personalization.

Cultural Adaptation Limitations: While our knowledge graph included 30 cultural practice nodes, these represent only the most common practices across Uganda's major ethnic groups. Regional variations in food preparation methods, religious dietary restrictions, and traditional food combinations are not fully captured. This limitation is most pronounced in Northern Uganda, where our training data is sparser due to logistical challenges in data collection.

Temporal Dynamics: The system currently uses static seasonal availability data based on historical patterns. Climate change and market disruption scan alter food availability in ways not reflected in our knowledge graph. We plan to address this through integration with real-time market price APIs, but such data sources are not yet available for rural Uganda.

Evaluation Limitations: Our evaluation relied on link prediction accuracy as a proxy for recommendation quality, but we have not yet conducted clinical trials to measure actual health outcomes. While high AUC suggests good discrimination between appropriate and inappropriate foods, the ultimate measure of success is improvement in nutritional status and health outcomes for elderly users.

XIV. CONCLUSIONS AND FUTURE WORK

This work has successfully demonstrated that a graph enhanced artificial intelligence paradigm provides a viable, powerful and practically deployable approach for addressing the complex challenge of the elderly malnutrition in low resource setting. The strategic integration of a meticulously constructed, culturally grounded Knowledge Graph with a sophisticated ensemble of Graph Neural Networks has enabled the creation of a system capable of delivering personalized, evidence based and culturally attuned nutritional guidance at scale. Our key contributions include: (1) the first large-scale digitized Ugandan Food Knowledge Graph, (2) a novel ensemble architecture combining compositional reasoning, heterogeneous graph modeling, and attention mechanisms, and (3) empirical demonstration that graph-based approaches outperform traditional machine learning for nutrition recommendation, achieving 0.92 AUC compared to 0.65 for baseline methods.

The system's deployment as a production Progressive Web Application with 15ms inference time demonstrates the practical feasibility of deploying advanced AI in resource constrained environments. The multi-language voice interface and offline caching capabilities address key accessibility barriers, while the ensemble architecture provides robust performance across diverse user profiles.

Future Research Directions

Wearable Device Integration We plan to integrate real time health data from wearable devices including heart rate monitors and continuous glucose monitors. This integration will enable dynamic recommendation adjustment based on current physiological state, moving from static meal plans to adaptive nutrition guidance that responds to real-time health indicators.

Large Language Model Augmentation: Future work will explore using GNN embeddings to prompt local Large Language Models for natural language explanation generation. This approach would enable the system to explain recommendations in culturally appropriate language, improving user trust and adherence. We are particularly interested in fine-tuning Llama-3 on Ugandan nutrition texts to create a culturally-aware explanation system.

Clinical Validation: We are planning a future randomized controlled trial with 200 elderly participants to measure the system's impact on health outcomes including BMI, hemoglobin levels, and dietary diversity scores. This trial will provide crucial evidence for the system's clinical effectiveness and inform future improvements.

Geographic Expansion: We aim to extend the knowledge graph to cover additional East African countries including Kenya, Tanzania, and Rwanda. This expansion will require collecting region specific food composition data and adapting the cultural practice nodes to reflect local traditions.

Explainable AI: Future versions will incorporate explainability techniques such as GNNExplainer to provide users with transparent reasoning about why specific foods are recommended. This transparency is crucial for building trust and enabling users to make informed decisions about their nutrition.

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